Exploratory Data Analysis

PROJECT 2 | E-COMMERCE ORDER DATA

Executive summary

This analysis examines 1,200 supplied order records to describe typical order values, assess the distribution across products and transaction categories, investigate time patterns, and flag unusually large orders. The workflow uses a cleaned Power BI dataset produced in Project 1 and documented descriptive statistics, charts, and an interquartile-range (IQR) outlier test.

Headline indicator	Verified result
Orders	1,200
Sum of recorded order values	1,264,761.96
Mean / median order value	1,053.97 / 823.62
Order value range	11.39 to 3,456.40
Potential IQR outliers	8 (0.67% of orders)

Objective and scope

DecodeLabs Project 2 requires descriptive statistics (mean, median, count), trends and outliers, and a written synthesis of observations. This report expands those requirements with categorical distributions and a documented IQR rule. Figures represent recorded order values, not confirmed realized revenue: the data includes cancelled, returned, and pending orders.

Data and methodology

Source: Dataset for Data Analytics (1).xlsx, Sheet1, 1,200 rows and 14 original fields. The Power BI EDA project uses the cleaned Project 1 version with proper Date typing, standardized selected text fields, and missing coupon codes labeled “No Coupon” under a documented assumption. The cleaning process retained all 1,200 orders. Two arithmetic audit fields were also created to compare order totals with Quantity × UnitPrice.

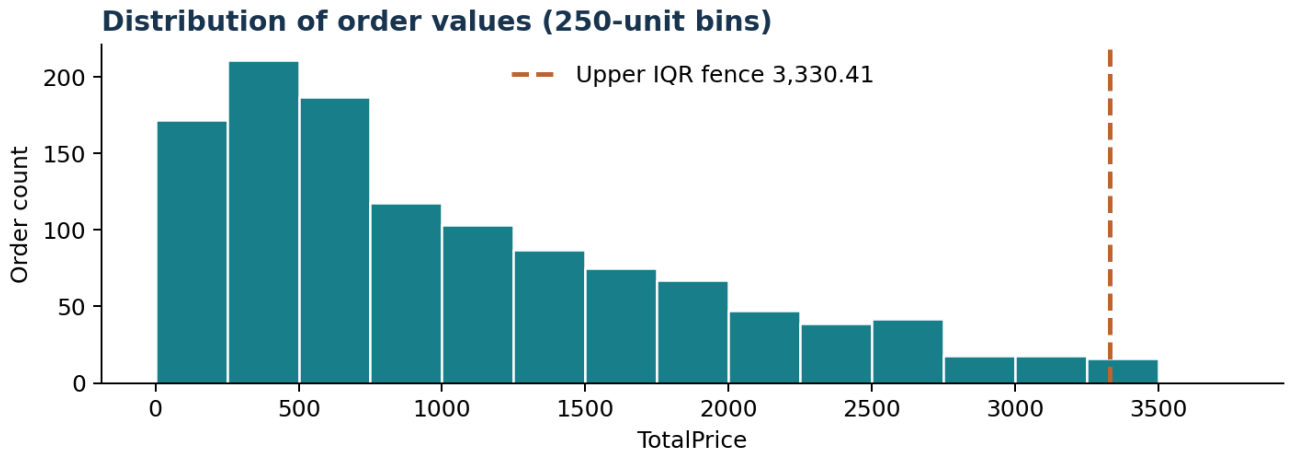
Analytical methods: distinct order counts, mean and median of TotalPrice, grouped sums and counts, monthly aggregation, frequency distribution, and Tukey’s 1.5 × IQR fences. Data summaries below were independently recomputed from the uploaded Excel source to verify the reported values.

01 Descriptive statistics

Statistic	TotalPrice result
Count	1,200
Mean	1,053.97
Median	823.62
Minimum	11.39
Q1 (25th percentile)	410.52
Q3 (75th percentile)	1,578.47
Maximum	3,456.40
Standard deviation	819.86

Interpretation: The mean exceeds the median by 230.35, consistent with a distribution pulled upward by a smaller number of high-value transactions. The standard deviation indicates substantial variation in order sizes.

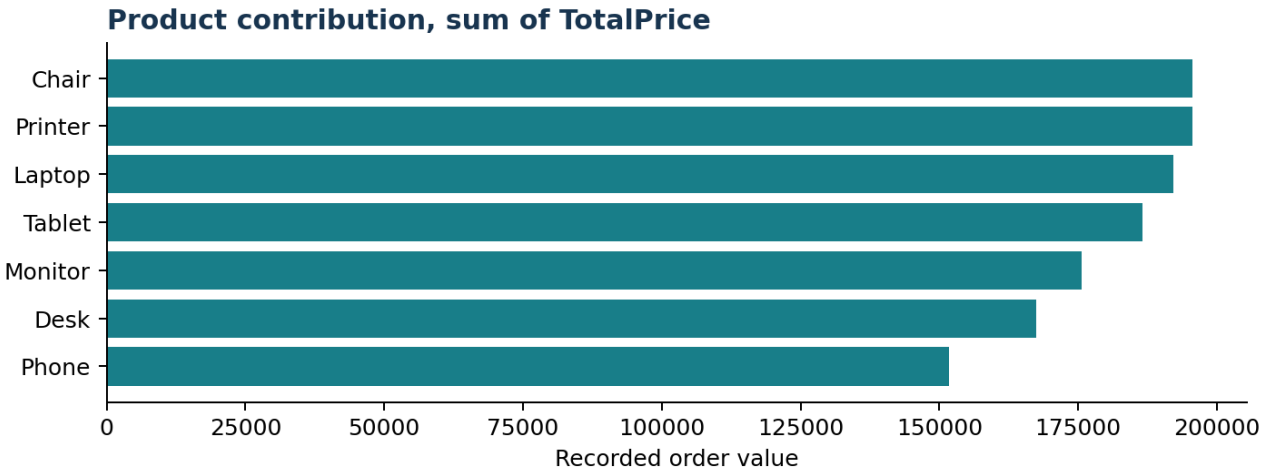
Order value distribution



The distribution is right-skewed: frequencies are concentrated toward smaller and mid-sized orders, with a progressively thinner high-value tail. The dashed reference line marks the IQR upper fence used for outlier screening.

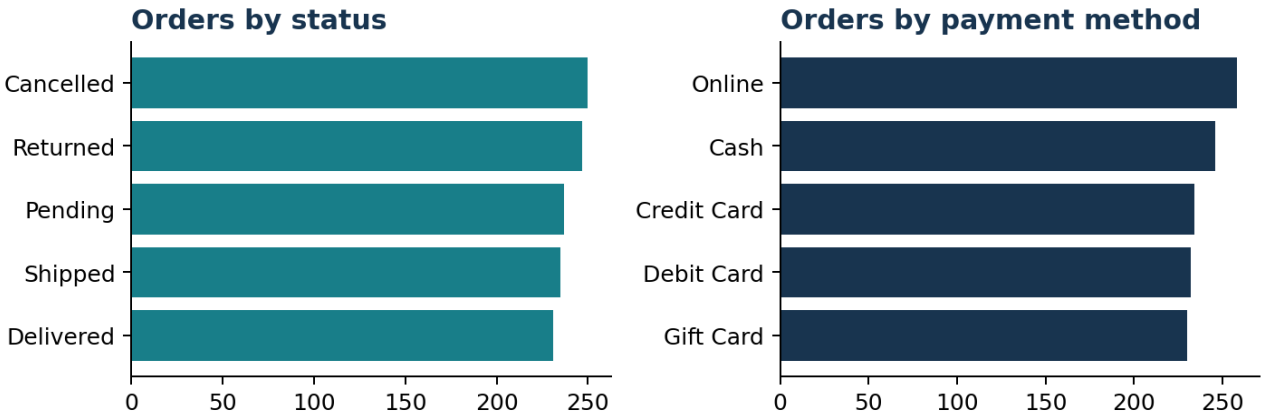
02 Product and transaction patterns

Product analysis



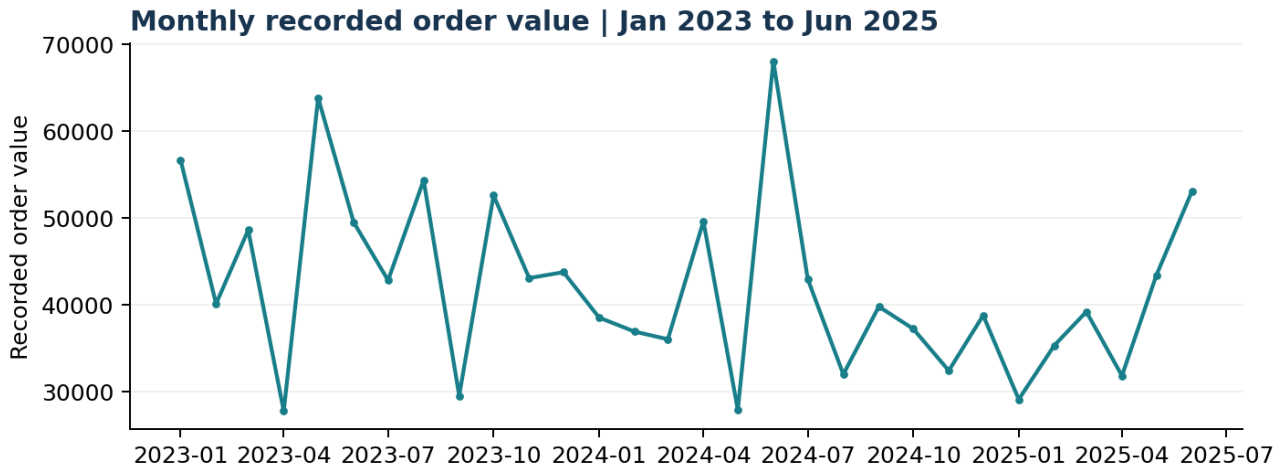
Chair (195,620.11) and Printer (195,612.61) are virtually tied for the highest sum of recorded order values; Phone is lowest (151,722.39). These differences reflect both order counts and order sizes. They should not be interpreted as profit or fulfilled sales.

Order status and payment method



The five statuses are relatively balanced: Cancelled 250, Returned 247, Pending 237, Shipped 235, and Delivered 231. Online is the most frequent payment method (258 orders), followed by Cash (246), Credit Card (234), Debit Card (232), and Gift Card (230).

03 Time trends and coverage



Year	Orders	Recorded order value
2023	510	552,643.24
2024	459	480,235.87
2025	231	231,882.85

Coverage runs from 1 January 2023 through 30 June 2025. The 2025 total covers only six months and is therefore not directly comparable with the full 2023 and 2024 totals. The highest recorded month was June 2024 (68,068.54). Month-to-month variation warrants follow-up by order counts and statuses before any claim about demand growth or seasonality.

Interpreting the trend responsibly

A simple yearly line chart can exaggerate decline if a partial year is treated as complete. A monthly view, or comparison of like-for-like January–June periods, is more appropriate. No causal claims are made because the dataset provides no promotion calendar, seasonality controls, or confirmed payment settlement information.

04 IQR outlier analysis

The IQR method identifies possible unusual transactions, not necessarily data-entry errors. Under the 1.5 × IQR rule, any TotalPrice below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR is flagged.

Parameter	Value
Q1	410.52
Q3	1,578.47
IQR	1,167.95
Lower fence	-1,341.41
Upper fence	3,330.41
Flagged orders	8 of 1200 (0.67%)

Orders above the upper fence

OrderID	Date	Product	TotalPrice
ORD200789	2023-08-17	Tablet	3,456.40
ORD201122	2023-06-07	Monitor	3,390.95
ORD200632	2023-05-02	Laptop	3,390.80
ORD200469	2023-11-26	Chair	3,384.90
ORD200328	2023-02-28	Tablet	3,370.20
ORD200107	2023-03-27	Printer	3,353.75
ORD200326	2024-07-01	Laptop	3,352.40
ORD201065	2023-10-30	Printer	3,334.00

All eight flagged values lie above the upper fence; none fall below the negative lower fence. Their amounts range from 3,334.00 to 3,456.40. Review order composition, pricing rules, and transaction status before excluding any record. For EDA, retain the outliers and disclose their influence on summary statistics.

05 Findings, limits and recommendations

Key findings

- Order values are positively skewed. Mean order value (1,053.97) exceeds the median (823.62).
- Product recorded order value is highest for Chair and Printer and lowest for Phone.
- Online payment has the most orders (258), but payment methods are broadly balanced.
- Monthly recorded order values fluctuate; the 2025 observation window ends in June.
- Eight orders (0.67%) exceed the upper 1.5 × IQR fence and merit review, not automatic deletion.

Limitations

- The 14-column source does not provide profit, tax, refunds, settlement status, or shipping charges. TotalPrice cannot be equated to realized revenue.
- The 2025 period is incomplete. Annual year-over-year comparisons would be misleading without aligning observation windows.
- OrderStatus is a snapshot without change history; cancelled and returned rows are included in recorded totals.
- The coupon null replacement uses a reporting label; it does not prove that no coupon was used.
- The report analyzes provided order records and does not establish causal drivers of product, payment, or seasonal variation.

Recommended follow-up

1. Segment product trends by delivered or completed status before making business revenue recommendations.
2. Compare matched months across years and include order counts alongside total values.
3. Review the eight flagged orders against pricing and transaction records.
4. Preserve the original dataset, Power Query transformation history, and Power BI measures for reproducibility.

Project deliverables and reproducibility

Primary deliverable: DecodeLabs_Project2_EDA.pbix with five report pages covering descriptive statistics, product contribution, time trends, status/payment distributions, and order value distribution with outlier details. Supporting file: this EDA report. Source workbook: Dataset for Data Analytics (1).xlsx. Open the .pbix file in Power BI Desktop; if refresh is required, update the source workbook path in Power Query to the workbook's local location.

Source basis: DecodeLabs "Data Analytics Project 2, Industrial Training Kit," page 4; uploaded Project 2 Excel dataset; and the Power BI work reviewed in this conversation. All statistics and the eight order identifiers shown in this report were checked against the uploaded source workbook.